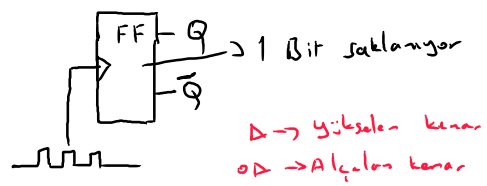


Flip - Floplar

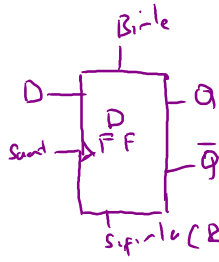


-D FF
-JK FF
-SR FF
-T FF

D Flip Flopu

Birle (SET)

(t)



$$Q(t+1) = D(t)$$

$$\left. \begin{aligned} Q(t+1) &= Y \\ Q(t) &= y \end{aligned} \right\} \rightarrow \text{ile gösterilir}$$

giriş	saat	çıkışlar
D		Q, Q̄
0	[clock signal]	0, 1
1	[clock signal]	1, 0

Bir sonraki çıkış

D	0	1
0	0	1
1	0	1

$y = Q(t+1)$

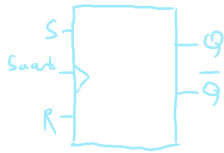
$$y = D$$

$$Q(t) Q(t+1)$$

y	00	01	11	10
D	0	1	1	0

Uyarma tablosu

Set/Reset (SR) FF



$$Q(t+1) = Q(t) \cdot R(t) + S(t)$$

$$R(t) \cdot S(t) = 0 \text{ için tanımlı}$$

$$R(t) \cdot S(t) = 1 \text{ için tanımsız}$$

giriş	saat	çıkışlar
S, R		Q, Q̄
0, 0	[clock signal]	aynı kalır
0, 1	[clock signal]	0, 1
1, 0	[clock signal]	1, 0
1, 1	[clock signal]	tanımsız

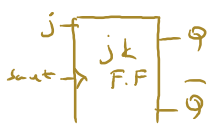
Çıkmı durumu

$y = \text{giriş di}$
 $y = \text{sonraki}$

S	R	0	1	1	0
SR	0	0	1	0	1

$y = S + R' y$

J K F.F. $\rightarrow$ SR'in 11'inin tanımlı hali



$$Q(t+1) = y$$

$$y = y \cdot k(t) + \bar{y} \cdot j(t)$$

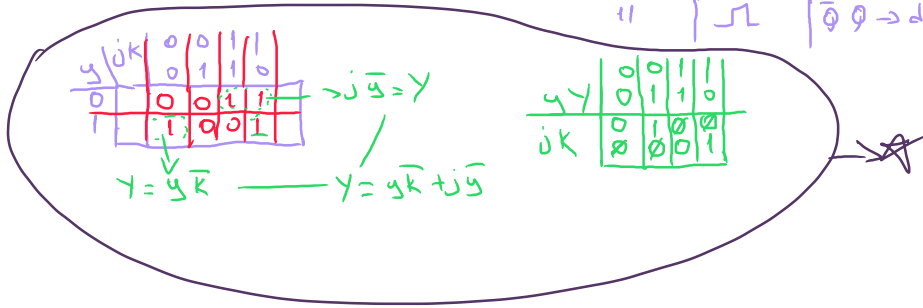
giriş	saat	çıkış
J, K		Q, Q̄
0, 0	[clock signal]	aynı kalır
0, 1	[clock signal]	0, 1
1, 0	[clock signal]	1, 0
1, 1	[clock signal]	0, 1

sonraki (t+1)

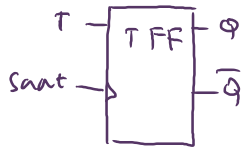
←

0 1
1 0
1 1

0 1
1 0
0 0 → deşillendir



TFF [Toggle (tersine çevirme)]



$$Q(t+1) = Q(t) \oplus T(t)$$

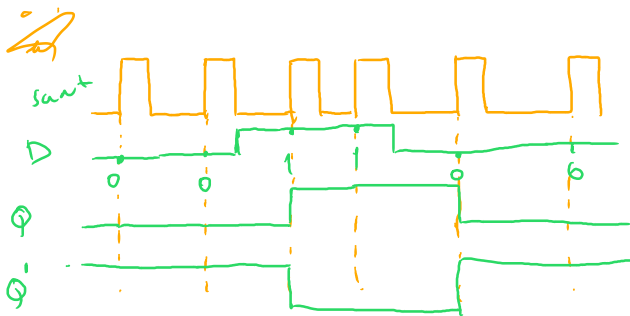
0 0 1 1
1 0 1 0
0 1 0 1

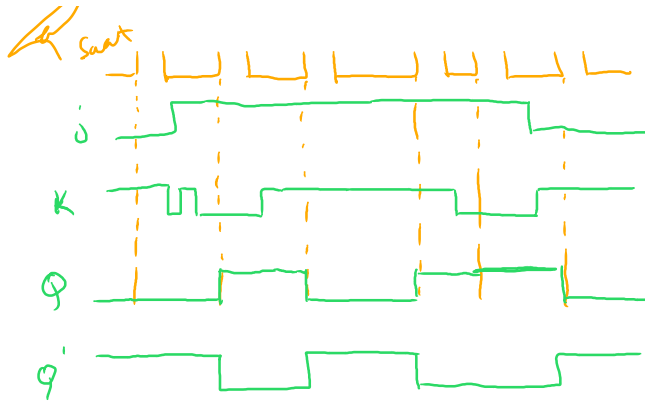
giriş T	S	çıkış Q Q̄
0	0	Q Q̄ → aynı
1	1	Q̄ Q → ters

Y	T	0	1
0	0	0	1
1	1	1	0

$$Y = TQ' + T'Q$$

$$Y = T \oplus Q$$





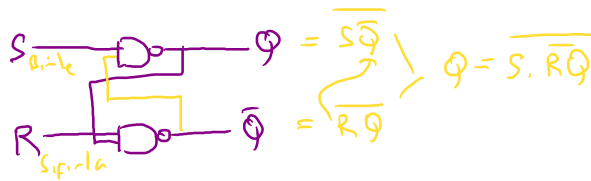
JK FF ile D ve T tipi FF yapılabilir mi?



Ana Uydu FF



Tutucular



$Q(t)=1$ ise

$SR=00$ $Q(t+1)=\text{tanımsız}$

$SR=01$ $Q(t+1)=0$

$SR=10$ $Q(t+1)=1$

$SR=11$ $Q(t+1)=Q(t)=1$

$Q(t)=0$ ise

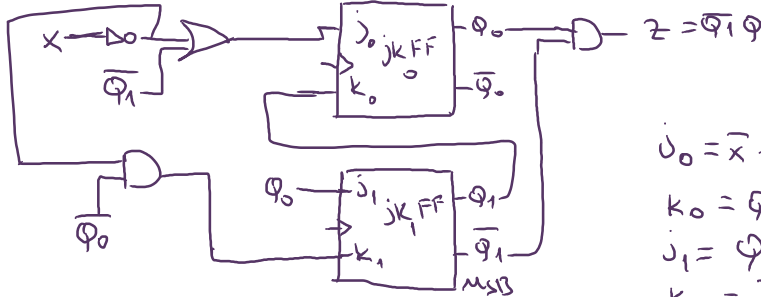
$SR=00$ $Q(t+1)=\text{tanımsız}$

$SR=01$ $Q(t+1)=0$

$SR=10$ $Q(t+1)=1$

$SR=11$ $Q(t+1)=0=Q(t)$

2



$$J_0 = \bar{x} + \bar{Q}_1$$

$$K_0 = Q_1$$

$$J_1 = Q_0$$

$$K_1 = \bar{x} \cdot \bar{Q}_0$$

F F girişleri

durumlar	y (k)		x=0				x=1				x=0				x=1				Z
	Q ₁	Q ₀	J ₀	K ₀	J ₁	K ₁	J ₀	K ₀	J ₁	K ₁	J ₀	K ₀	J ₁	K ₁	J ₀	K ₀			
d ₀	0	0	0	1	1	0	0	1	0	1	0	1	0	1	0	1	0		
d ₁	0	1	1	0	1	0	1	0	1	0	1	1	0	1	1	1	1		
d ₂	1	0	0	1	1	0	0	0	1	0	1	1	0	1	1	0	0		
d ₃	1	1	1	0	1	1	0	0	1	1	0	1	1	0	1	0	0		

y	x0	x1	Z
d0	d1	d1	0
d1	d3	d3	1
d2	d1	d2	0
d3	d2	d2	0

